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Semiconductor chip having stepped conductive pillars

Abstract

In an implementation, a semiconductor chip includes a device layer, an interconnect layer fabricated on the device layer, the interconnect layer including a conductive pad, and a conductive pillar coupled to the conductive pad. The conductive pillar includes at least a first portion having a first width and a second portion having a second width, the first portion being disposed between the second portion and the conductive pad, wherein the first width of the first portion is greater than the second width of the second portion.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This is a continuation application of and claims priority for patent entitled to a filing date and claiming the benefit of earlier-filed U.S. patent application Ser. No. 17/528,523, filed Nov. 17, 2021. Each patent application cited herein is hereby incorporated by reference in its entirety.

BACKGROUND

(1) Generally, semiconductor chips comprise active devices (e.g., transistors, capacitors, etc.), and an interconnect layer forming connections to the active devices, and input/output (I/O) contacts to provide signal pathways, power, and ground for the interconnection layers and active devices. The interconnect layer generally includes dielectric layers and metal layers that provide all of the required connections between the active devices and the I/O contacts (and between individual active devices). These dielectric layers can be formed from extremely low-k (ELK) dielectric materials with dielectric constants (k-value) less than 3. The ELK dielectric materials provide

several advantages, including reduced parasitic capacitance, faster switching speeds, and lower heat dissipation compared to other conventional dielectric materials such as silicon dioxide (SiO₂).

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 sets forth a sectional view of an example semiconductor chip having stepped conductive pillars according to some implementations of the present disclosure.
- (2) FIG. 2 sets forth a detailed sectional view of the semiconductor chip of FIG. 1 according to some implementations.
- (3) FIG. 3 sets forth a sectional view of a portion of an example process flow for fabricating a semiconductor chip having stepped conductive pillars according to some implementations.
- (4) FIG. 4 is another portion of the example process flow for fabricating a semiconductor chip having stepped conductive pillars according to some implementations.
- (5) FIG. 5 is another portion of the example process flow for fabricating a semiconductor chip having stepped conductive pillars according to some implementations.
- (6) FIG. 6 is another portion of the example process flow for fabricating a semiconductor chip having stepped conductive pillars according to some implementations.
- (7) FIG. 7 is another portion of the example process flow for fabricating a semiconductor chip having stepped conductive pillars according to some implementations.
- (8) FIG. 8 is another portion of the example process flow for fabricating a semiconductor chip having stepped conductive pillars according to some implementations.
- (9) FIG. 9 is another portion of the example process flow for fabricating a semiconductor chip having stepped conductive pillars according to some implementations.
- (10) FIG. 10 is another portion of the example process flow for fabricating a semiconductor chip having stepped conductive pillars according to some implementations.
- (11) FIG. 11 is another portion of the example process flow for fabricating a semiconductor chip having stepped conductive pillars according to some implementations.
- (12) FIG. 12 is another portion of the example process flow for fabricating a semiconductor chip having stepped conductive pillars according to some implementations.
- (13) FIG. 13 is another portion of the example process flow for fabricating a semiconductor chip having stepped conductive pillars according to some implementations.
- (14) FIG. 14 is another portion of the example process flow for fabricating a semiconductor chip having stepped conductive pillars according to some implementations.
- (15) FIG. 15 sets forth a detailed sectional view of another implementation of a semiconductor chip having stepped conductive pillars according to some implementations.
- (16) FIG. 16 sets forth a detailed sectional view of another implementation of a semiconductor chip having stepped conductive pillars according to some implementations.
- (17) FIG. 17 sets forth a detailed sectional view of another implementation of a semiconductor chip having stepped conductive pillars according to some implementations.
- (18) FIG. 18A sets forth a perspective view of an example shape of a stepped conductive pillar according to some implementations.
- (19) FIG. 18B sets forth a perspective view of an example shape of a stepped octagonal conductive pillar according to some implementations.
- (20) FIG. 18C sets forth a perspective view of an example shape of a stepped and offset conductive pillar according to some implementations.
- (21) FIG. 19 sets forth a perspective view of another example shape of a stepped conductive pillar according to some implementations.
- (22) FIG. 20 sets forth a perspective view of yet another example shape of a stepped conductive

pillar according to some implementations

(23) FIG. 21 sets forth a flowchart of an example method of fabricating a semiconductor chip having stepped conductive pillars according to some implementations.

(24) FIG. 22 sets forth an additional flowchart of the example method of fabricating a semiconductor chip having stepped conductive pillars according to some implementations.

(25) FIG. 23 sets forth an additional flowchart of the example method of fabricating a semiconductor chip having stepped conductive pillars according to some implementations.

DETAILED DESCRIPTION

(26) While ELK materials may be used to improve the electrical characteristics of the metallization layers and thereby increase the overall speed or efficiency of the semiconductor device, these materials have a significant structural drawback. With the reduction in the k-value (i.e., the dielectric constant value), the hardness and mechanical modulus of the ELK material are reduced, resulting ELK strength reduction. Thus, ELK materials are less capable than other dielectric materials (e.g., SiO₂) in handling the stresses applied to them in the semiconductor package. For example, chip-package interactions between the semiconductor chip and the substrate in the semiconductor package can cause delamination or cracking in the semiconductor chip.

(27) Furthermore, with advancements in nanoscale wafer technology, increasing chip performance drives the requirements for a finer pitch between interconnects (e.g., solder bumps, conductive pillars or posts) of the semiconductor chip. In some solutions, the fine pitch between interconnects is achieved by reducing the size of the under-bump metallization (UBM) layer. However, this can further increase stress on the ELK material, causing cracking or delamination. In other solutions, the size of the UBM is increased to mitigate against ELK stress. However, this increases the size of the solder bump, which can lead to solder bridging among fine pitched interconnects.

(28) To that end, various implementations of a semiconductor chip having stepped conductive pillars are described in this specification. The semiconductor chip includes a device layer and an interconnect layer fabricated on the device layer, with the interconnect layer including a conductive pad. The semiconductor chip also includes a conductive pillar coupled to the conductive pad. The conductive pillar includes at least a first portion having a first width and a second portion having a second width, with the first portion being disposed between the second portion and the conductive pad. The first width of the first portion is greater than the second width of the second portion. In some implementations, the semiconductor chip further includes a solder cap on an end of the second portion. In some implementations, the interconnect layer includes extremely low-k (ELK) dielectric material.

(29) In some implementations, the conductive pillar forms a stepped cylinder. In other implementations, the conductive pillar forms a stepped cuboid.

(30) In some implementations, the conductive pillar includes a third portion disposed between the first portion and the second portion, the third portion having a third width that is smaller than the second width.

(31) In some implementations, the semiconductor chip includes a passivation layer formed on the interconnect layer, a polymer layer formed on the passivation layer, an aperture through the passivation layer and the polymer layer where the aperture exposes a portion of the conductive pad, and an under-bump metallization layer formed on at least the exposed portion of the conductive pad and coupling the conductive pillar to the conductive pad.

(32) In some implementations, the semiconductor chip includes a passivation layer formed on the interconnect layer, an aperture in the passivation layer, where the aperture exposes a portion of the conductive pad, an under-bump metallization layer formed on at least the exposed portion of the conductive pad and coupling the conductive pillar to the conductive pad, and a polymer layer formed on the first portion of the conductive pillar and the under-bump metallization layer.

(33) In some implementations, the semiconductor chip includes a passivation layer formed on the interconnect layer, a first polymer layer formed on the passivation layer, an aperture in the

passivation layer and the first polymer layer, where the aperture exposes a portion of the conductive pad, an under-bump metallization layer formed on at least the exposed portion of the conductive pad and coupling the conductive pillar to the conductive pad, and a second polymer layer formed on the first portion of the conductive pillar and the under-bump metallization layer.

(34) A variation of the embodiment is directed to a semiconductor chip package including stepped conductive pillars. The semiconductor chip package includes a package substrate including a bond pad and a semiconductor chip mounted on the package substrate. The semiconductor chip includes a device layer and an interconnect layer fabricated on the device layer, where the interconnect layer includes a conductive pad. The semiconductor chip also includes a conductive pillar interconnecting the conductive pad and the bond pad of the package substrate. The conductive pillar includes at least a first portion having a first width and a second portion having a second width. The first portion is disposed between the second portion and the conductive pad, and the second portion is disposed between the first portion and the package substrate, where the first width of the first portion is greater than the second width of the second portion. In some implementations, the semiconductor chip also includes a solder cap on an end of the second portion. In some implementations, the interconnect layer includes extremely low-k (ELK) dielectric material. In some implementations of the semiconductor chip package, the conductive pillar forms a stepped cylinder.

(35) In some implementations, the conductive pillar includes a third portion disposed between the first portion and the second portion, the third portion having a third width that is smaller than the second width.

(36) In some implementations, the semiconductor chip in the package includes a passivation layer formed on the interconnect layer, a polymer layer formed on the passivation layer, an aperture through the passivation layer and the polymer layer, the aperture exposing a portion of the conductive pad, and an under-bump metallization layer formed on at least the exposed portion of the conductive pad and coupling the conductive pillar to the conductive pad.

(37) In some implementations, the semiconductor chip in the package includes a passivation layer formed on the interconnect layer, an aperture in the passivation layer, the aperture exposing a portion of the conductive pad, an under-bump metallization layer formed on at least the exposed portion of the conductive pad and coupling the conductive pillar to the conductive pad, and a polymer layer formed on the first portion of the conductive pillar and the under-bump metallization layer.

(38) In some implementations, the semiconductor chip in the package includes a passivation layer formed on the interconnect layer, a first polymer layer formed on the passivation layer, an aperture in the passivation layer and the first polymer layer, the aperture exposing a portion of the conductive pad, an under-bump metallization layer formed on at least the exposed portion of the conductive pad and coupling the conductive pillar to the conductive pad, and a second polymer layer formed on the first portion of the conductive pillar and the under-bump metallization layer.

(39) Another variation of the embodiment is directed to a method of fabricating a semiconductor device having stepped conductive pillars. The method includes providing a semiconductor chip substrate that includes a conductive pad and a passivation layer through which the conductive pad is at least partially exposed. The method also includes fabricating an under-bump metallization layer over an exposed portion of the conductive pad. The method further includes fabricating a first portion of a conductive pillar over the conductive pad, where the first portion of the conductive pillar has a first width. The method further includes fabricating a second portion of the conductive pillar on the first portion, where the second portion has a second width that is smaller than the first width.

(40) In some implementations, fabricating a first portion of the conductive pillar over the conductive pad, where the first portion of the conductive pillar has a first width, includes dispensing a first layer of photoresist material, creating a first cavity in the first layer of photoresist

material by photolithography, and forming the first portion of the conductive pillar in the first cavity. In these implementations fabricating a second portion of the conductive pillar on the first portion, where the second portion has a second width that is smaller than the first width, includes dispensing a second layer of photoresist material, creating a second cavity in the second layer of photoresist material by photolithography, the second cavity having a smaller width than the first cavity, forming the second portion of the conductive pillar in the second cavity.

(41) Implementations in accordance with the present disclosure will be described in further detail beginning with FIG. 1. Like reference numerals refer to like elements throughout the specification and drawings. FIG. 1 sets forth a sectional view of an example semiconductor chip **100** in accordance with some implementations of the present disclosure. Implementations of the semiconductor chip **100** are useful in high performance applications, such as a personal computer, a notebook, a tablet, a smart phone, a data center, or applications involving large scale databases and/or analytics including finance, life sciences, and/or artificial intelligence. It will be appreciated that many other applications are possible.

(42) In the example of FIG. 1, the semiconductor chip **100** includes a device layer **106**. The device layer **106** includes layers of metallization and dielectric material implementing a variety of integrated circuits that are not depicted for clarity. The semiconductor chip **100** also includes an interconnect layer **104** that include layers of metallization and dielectric material, some of which are not depicted here for clarity. The layers of metallization include conductive structures, such as traces, pads, and vias, that provide routing for power, ground, and input/output ('I/O') signals between external interconnects and components in the device layer **106** and among components in the device layer **106**. A final layer of metallization in the interconnect layer **104** includes conductive pads **102**.

(43) The conductive pads **102** provide a bond site for external interconnect structures. External interconnect structures such as the conductive pillars **120** are coupled to the exposed portions of the conductive pad **102**. The conductive pillars **120** include at least a base portion **122** and an end portion **124**. In some variations, the conductive pillars **120** include multiple cylindrical portions **122**, **124**. In these examples, the width of the base portion **122** is greater than the width of the end portion **124**. In other variations, the conductive pillars **120** include multiple portions **122**, **124** that are not cylindrical (e.g., cuboid). In these examples, the largest width of the base portion **122** is greater than the largest width of the end portion **124**. In some examples, as depicted, a solder cap **126** is disposed on a planar surface of the end portion **124**. The solder cap **126** can be composed of a well-known solder material.

(44) As discussed above, it is not desirable to increase the pitch of adjacent solder caps **126** to minimize the risk of solder bridging. To that end, the width of the end portion should not be increased beyond design dimensions. To mitigate the impact of stress on the underlying interconnect layer **104** and to reduce the possibility of cracking and delamination in the interconnect layer **104**, the width of the base portion of each conductive pillar **120** is selected to be greater than a particular design requirement for the width of the end portion **124** (e.g., a design requirement for interconnect pitch that necessitates a particular end portion width **224**). The greater width of the base portion allows for a greater width of an underbump metallization layer (described in greater detail with respect to FIG. 2) thus providing ELK stress reduction.

(45) For further explanation, FIG. 2 sets forth a detailed view of a portion of the semiconductor chip **100** identified by the dashed rectangle in FIG. 1. To facilitate explanation, the orientation of the semiconductor chip **100** in FIG. 2 is flipped with respect to FIG. 1. In the example of FIG. 2, the interconnect layer **104** (e.g., a back-end-of-line (BEOL) interconnect structure) includes multiple layers of dielectric material including a final dielectric layer supporting the conductive pad **102**. In some implementations, the dielectric material is an ELK dielectric. The conductive pad **102** can be composed of a suitable metal or other conductive material. In some implementations, the conductive pad **102** is an aluminum pad or a copper pad.

(46) A passivation layer **204** is formed over the final dielectric layer of the interconnect layer **104**. In some variations, as shown, the passivation layer **204** overlaps outer portions of the conductive pad **102**. In other variations, the passivation layer **204** abuts the conductive pad **102** without overlap. The passivation layer **204** includes a passivation aperture **206** through which at least a portion of the conductive pad **102** is exposed.

(47) A polymer layer **208** is formed over the passivation layer **204**. In some examples, the polymer layer **208** is a polyimide layer. In some implementations, as shown in FIG. 2, the polymer layer **208** extends partially into the passivation aperture **206** and over outer portions of the conductive pad **102**. Thus, in these examples, the passivation layer **204** is interposed between the polymer layer **208** and the conductive pad **102**. The polymer layer **208** also defines a polymer aperture **210**, which exposes the underlying the conductive pad **102**. In some examples, as shown, the polymer aperture **210** is smaller than the passivation aperture **206**. In other variations, the polymer aperture **210** is greater than the passivation aperture **206**. In still further variations, the polymer aperture **210** and the passivation aperture **206** are coterminous.

(48) An under-bump metallization (UBM) layer **212** is formed on the exposed portion of the conductive pad **102** through the passivation aperture **206** and the polymer aperture **210**. In some implementations, as shown in FIG. 2, the UBM layer **212** contacts and overlaps portions of the polymer layer **208** adjacent to the polymer aperture **210**. The UBM layer **212** is electrically coupled to the conductive pad **102**. In various implementations, the UBM layer **212** can be formed from titanium (Ti), titanium tungsten (TiW), and the like. In an implementation, the UBM layer **212** is formed by an adhesion layer of titanium that also acts as diffusion barrier. As shown, the UBM layer **212** defines a UBM width that is equal to the base portion width **222**.

(49) The conductive pillar **120** is disposed on the UBM layer **212**. The conductive pillar can be formed from copper or another suitable material. In one example, the conductive pillar is formed from copper. The conductive pillar **120** has a shape that includes different widths or diameters along an axis of the conductive pillar **120** perpendicular to the interconnect layer **104**. In the example of FIG. 2, a base portion **122** of the conductive pillar **120** proximate to the UBM layer **212** includes sidewalls that define a base portion width **222**. In some examples, the base portion width **222** is equal to the UBM layer width. An end portion **124** of the conductive pillar distal to the UBM layer **212** includes sidewalls that define an end portion width **224**. In the example of FIG. 2, the conductive pillar is a stepped cylinder with the base portion **122** having a first diameter and the end portion having a second diameter. Where the conductive pillar is an elliptical cylinder having a major axis and a minor axis, the width of each portion is, respectively, the width along the major axis or the widest distance between the sidewalls. Where the base portion **122** and the end portion **124** are not cylindrical (e.g., cuboid), the width of the base portion **122** and the end portion **124** is the largest width between respective lateral faces. When present, the solder cap **126** is formed on the planar surface of the end portion **124**, farthest from the interconnect layer **104**. In some implementations, a diffusion barrier layer (not shown) is formed between the solder cap **126** and the end portion **124**. The diffusion barrier layer can be formed from titanium (Ti), titanium tungsten (TiW), and the like.

(50) As discussed above, increasing the pitch of adjacent solder caps to reduce risk of solder bridging is undesirable. To maintain the pitch, the end portion width **224** should not be increased beyond design dimensions. To mitigate the impact of stress on the underlying interconnect layer **104** and to reduce the possibility of cracking and delamination in the interconnect layer **104** (while maintaining the end portion width **224**), the base portion width **222** is selected to be greater than the design requirement for the end portion width **224**. The greater base portion width **222** allows for a greater width of the UBM layer thus providing ELK stress reduction.

(51) For further explanation, an example process flow for fabricating the stepped conductive pillar structures will now be described in conjunction with FIGS. 3-14 and initially with reference to FIG. 3. For clarity, the process flow depicted in FIGS. 3-14 is illustrated with respect to a single

conductive pillar. However, it will be appreciated that, during the process flow for fabricating a stepped conductive pillar structure, multiple stepped conductive pillars can be fabricated.

(52) FIG. 3 is a sectional view that depicts placement of the semiconductor chip substrate **300** on a carrier **302** that supports the semiconductor chip during a bumping process in which the conductive pillars are formed. The semiconductor chip substrate **300** is provided after a wafer-level fabrication process to create the semiconductor chip substrate **300**. As such, the semiconductor chip substrate includes the device layer **106** and the interconnect layer **104** formed over the device layer **106**.

(53) The interconnect layer **104** includes a final dielectric layer supporting a conductive pad **102**. The passivation layer **204** is formed over the final metal layer and, in the example of FIG. 3, over the periphery of the conductive pad **102**. The passivation aperture **206** exposes a portion of the conductive pad **102** not overlapped by the passivation layer **204**.

(54) FIG. 4 sets forth a sectional view that depicts deposition of the polymer layer **208**. The polymer layer can be composed of a variety of polymeric material. The polymer layer **208** defines the polymer aperture **210** that exposes a portion of the conductive pad **102**. In some implementations, the polymer aperture **210** is smaller than the passivation aperture **206** such that some polymer material contacts and overlaps a portion of the conductive pad **102**. In other variations, the polymer aperture **210** is larger than the passivation aperture **206** such that the polymer material does not contact the conductive pad. In still further variations, the polymer aperture **210** and the passivation aperture **206** are coterminous, such that the polymer material does not contact the conductive pad **102**.

(55) FIG. 5 sets forth a sectional view that depicts deposition of an adhesion layer **502** and a seed layer **504**. The adhesion layer **502** can be composed of a variety of suitable materials, including titanium (Ti), titanium tungsten (TiW), and the like, and can be deposited by a variety of well-known techniques. In some implementations, the adhesion layer **502** is composed of titanium (Ti) and deposited by a sputtering process. In these implementations, the adhesion layer **502** also acts as a diffusion barrier. The seed layer **504** can be composed of a variety of suitable materials, including copper (Cu), gold (Au), and the like, and can be deposited by a variety of well-known techniques. In some implementations, the seed layer **504** is composed of copper (Cu) and deposited by a sputtering process. The adhesion layer **502** will later form the UBM layer **212** after an etching process, as will be explained below.

(56) FIG. 6 sets forth a sectional view that depicts a first level of photoresist material **602** dispensed over the seed layer **504**. For example, the photoresist material **602** is applied by spin coating. A photomask **604** is placed on or above the photoresist material **602**, the photomask **604** having a mask aperture **606** that aligns with the conductive pad **102** (e.g., aligned along a vertical axis through the center of the mask aperture **606** and the center of the conductive pad **102**). In some examples the mask aperture **606** has a width that is equal to the eventual base portion width **222**. A portion of the photoresist material **602'** is exposed to radiation such as UV light through the mask aperture **606**.

(57) FIG. 7 sets forth a sectional view that depicts development of the photoresist material **602** including the portion of the photoresist material **602'**, previously shown in FIG. 6, that has undergone exposure. The portion of the photoresist material **602'** is removed through the developing process. The removal of photoresist material **602'** creates a cavity **702** in the photoresist material **602** that is used to form the base portion **122** of the conductive pillar **120**.

(58) FIG. 8 sets forth a sectional view that depicts fabrication of the base portion **122** of the conductive pillar **120**. Metal forming the base portion **122** is deposited on the seed layer **504** in the cavity **702** (previously shown in FIG. 7) through various well-known techniques. In some implementations, the metal is deposited by electroplating. In some examples, the metal is copper (Cu). At this stage, because the base portion **122** and the seed layer **504** are composed of the same metal, the metal of the seed layer **504** in the cavity **702** becomes integrated with the metal of the base portion **122**.

(59) FIG. 9 sets forth a sectional view that depicts a second level of photoresist material **902** dispensed on surfaces of the seed layer **504** and over the periphery of the base portion **122**. For example, the photoresist material **902** is applied by spin coating. A photomask **904** is placed on or above the photoresist material **902**, the photomask **904** having a mask aperture **906** that aligns with the conductive pad **102**. In the example of FIG. 9, the mask aperture is aligned along a vertical axis through the center of the mask aperture **906** and the center of the conductive pad **102**. In some implementations, the mask aperture can be offset relative the same vertical axes. In some examples the mask aperture **906** has a width that is equal to the end portion width **224**. A portion of the photoresist material **902'** is exposed to radiation such as UV light through the mask aperture **906**. In some variations, the second level of photoresist material **902** is dispensed without stripping the first level of photoresist material **602**.

(60) FIG. 10 sets forth a sectional view that depicts development of the photoresist material **902** including the portion of the photoresist material **902'** that has undergone exposure. The portion of the photoresist material **902'**, previously shown in FIG. 9, is removed through the developing process. The removal of photoresist material **902'** creates a cavity **1002** in the photoresist material **902** that is used to form the end portion **124** of the conductive pillar **120**.

(61) FIG. 11. sets forth a sectional view that depicts fabrication of the end portion **124** of the conductive pillar **120**. Metal forming the end portion **124** is deposited in the cavity **1002**, previously shown in FIG. 10, through various well-known techniques. The metal (e.g., copper) is the same metal that is used to form the base portion **122**. Thus, the deposition of the metal forming the end portion **124** is joined with the metal forming the base portion **122**. In some implementations, the metal is deposited by electroplating.

(62) FIG. 12 sets forth a sectional view that depicts removal of the photoresist material **902** previously shown in FIG. 11. The second level of photoresist material **902** can be removed through a well-known stripping process. As depicted in FIG. 12, removal of the photoresist material **902** exposes the conductive pillar **120** including the base portion **122** and the end portion **124**, the end portion having a smaller width **224** than the base portion width **222**. In alternative implementations where the first level of photoresist material **602** is not stripped before dispensing the second level of photoresist material **902**, the photoresist material **602**, **902** can be stripped together.

(63) FIG. 13 sets forth a sectional view that depicts removal of the adhesion layer **502** and the seed layer **504**, previously shown in FIG. 12, except for portions below the conductive pillar **120**. In some implementations, the adhesion layer **502** and the seed layer **504** surrounding the conductive pillar **120** are removed by chemical etching. The remaining portion of the adhesion layer **502** forms the UBM layer **212**.

(64) FIG. 14 sets forth a sectional view that depicts the formation of solder caps **126** on the end portion **124** of the conductive pillar **120**. In some implementations, solder material is deposited on the planar surface of the end portion **124**. In some examples, a diffusion barrier layer (not shown) is formed on the planar surface of the end portion **124** prior to depositing the solder material. The diffusion barrier layer can be composed of nickel (Ni), or the like. Finally, a solder reflow process is performed.

(65) The techniques described herein can be expanded to include stepped conductive pillar arrangements other than those depicted in FIGS. 1 and 2. For further explanation, FIG. 15 sets forth a detailed view of another example semiconductor chip **1500** having another example stepped conductive pillar **1520**. Like the example semiconductor chip **100** detailed in FIG. 2, the example semiconductor chip **1500** includes a device layer, an interconnect layer **104**, a conductive pad **102**, a passivation layer **204**, a polymer layer **208**, and a UBM layer **212**. The example semiconductor chip **1500** is different from the example semiconductor chip **100** in that the conductive pillar **1520** includes a middle portion **1526** disposed between the base portion **122** and the end portion **124**. In some implementations, the base portion **122**, the end portion **124**, and middle portion **1526** form a stepped cylinder. In these implementations, the middle portion width **1528** is smaller than both the

base portion width **222** and the end portion width **224**. In other implementations, the base portion **122**, the end portion **124**, and middle portion **1526** form a stepped cuboid (described below). In these implementations, the largest width of the middle portion is smaller than the largest width of the base portion **122** and smaller than the largest width of the end portion **124**. The example of FIG. **15** also includes the solder cap **126** on the end portion **124**.

(66) An example process flow for fabricating a semiconductor chip such as the semiconductor chip **1600** is similar to the process flow depicted in FIGS. **3-14** except that a third level of photoresist material is dispensed and exposed and another portion of the conductive post is formed, as described above with reference to FIGS. **9-12**. The middle portion **1526** is fabricated using the second level of photoresist material **902** as a mask, while the end portion **124** is fabricated using an additional third level of photoresist material (not depicted) as a mask.

(67) For further explanation, FIG. **16** sets forth a detailed view of another example semiconductor chip **1600**. The example semiconductor chip **1600** is different from the example semiconductor chip **100** in that a polymer layer **1608** is formed over the UBM layer **212** instead of on the passivation layer **204**, such that the polymer layer **1608** overlies the UBM layer **212** and the base portion **122** of the conductive pillar **120** as well as the passivation layer **204**. The example of FIG. **16** also includes the solder cap **126** on the end portion **124**.

(68) An example process flow for fabricating a semiconductor chip such as the semiconductor chip **1600** is similar to the process flow depicted in FIGS. **3-14** except that the deposition of the polymer layer **208** at the stage depicted in FIG. **4** is omitted. Additionally, after removal of the adhesion layer **502** and the seed layer **504** from around the conductive pillar **120** at the stage depicted in FIG. **13**, the polymer layer **1608** is deposited. The polymer layer **1608** can be composed of a variety of polymeric materials.

(69) For further explanation, FIG. **17** sets forth a detailed view of another example semiconductor chip **1700**. The example semiconductor chip **1700** is different from the example semiconductor chip **100** in that a second polymer layer **1708** is formed over the UBM layer **212**, such that the polymer layer **1708** overlies the polymer layer **208**, the passivation layer **204**, the UBM layer **212**, and the base portion **122** of the conductive pillar **120**. The example of FIG. **17** also includes the solder cap **126** on the end portion **124**.

(70) An example process flow for fabricating a semiconductor chip such as the semiconductor chip **1700** is similar to the process flow depicted in FIGS. **3-14** with the addition that, after removal of the adhesion layer **502** and the seed layer **504** from around the conductive pillar **120** at the stage depicted in FIG. **13**, the polymer layer **1708** is deposited. The polymer layer **1708** can be composed of a variety of polymeric materials.

(71) FIG. **18A** sets forth an illustration of an example stepped circular-cylindrical conductive pillar **1800** in accordance with some implementations. The base portion and end portion of the circular-cylindrical conductive pillar **1800** each include opposing identical circular faces separated by a cylindrical sidewall. As each face of the base portion and the end portion are circular, the conductive pillar has a base portion diameter and an end portion diameter, which are respectively the base portion width **1802** and the end portion width **1804**.

(72) FIG. **18B** sets forth an illustration of an example stepped octagonal conductive pillar **1806** in accordance with some implementations. The base portion and end portion of the octagonal conductive pillar **1806** each include opposing identical octagonal faces separated by an octagonal sidewall. As each face of the base portion and the end portion are octagonal, the conductive pillar has a base portion diameter and an end portion diameter, which are respectively the base portion width **1808** and the end portion width **1810**.

(73) FIG. **18C** sets forth an illustration of an example stepped and offset circular-cylindrical conductive pillar **1812** in accordance with some implementations. The base portion and end portion of the circular-cylindrical conductive pillar **1812** each include opposing identical circular faces separated by a cylindrical sidewall. As each face of the base portion and the end portion are

circular, the conductive pillar has a base portion diameter and an end portion diameter, which are respectively the base portion width **1814** and the end portion width **1816**. In addition, the end portion has an axis **1818** that is off-center with respect to the axis **1820** of the base portion.

(74) FIG. **19** sets forth an illustration of an example stepped elliptical-cylindrical conductive pillar **1900** in accordance with some implementations. The base portion and end portion of the elliptical-cylindrical conductive pillar **1900** each include opposing identical elliptical faces separated by a cylindrical sidewall. As each face of the base portion and of the end portion are ellipses, the respective faces each include a major axis and a minor axis, where the width of the base portion is considered the width along its major axis (i.e., the largest width) and the width of the end portion is considered the width along its major axis (i.e., the largest width). Thus, the conductive pillar has a base portion width **1902** and an end portion width **1904**.

(75) FIG. **20** sets forth an illustration of an example stepped cuboid conductive pillar **2000** in accordance with some implementations. The base portion and end portion of the stepped cuboid conductive pillar **2000** are cuboids that each include four rectangular (or square) lateral faces forming sidewalls. Here, the width of the base portion is considered the largest width between its lateral faces and the width of the end portion is considered the largest width between its lateral faces. Thus, the conductive pillar has a base portion width **2002** and an end portion width **2004**.

(76) FIG. **21** sets forth a sectional view of an example semiconductor package **2100** implementing stepped conductive pillar interconnects on a semiconductor chip in accordance with some implementations of the present disclosure. In the example of FIG. **21**, the semiconductor chip **100** of FIG. **1** is mounted on a substrate **2102** that includes multiple bond pads **2104** on a surface of the substrate. A solder mask **2106** is provided of the surface of the substrate **2102** and includes apertures at least partially exposing the bond pads **2104**. In some examples, the substrate **2102** includes a redistribution layer or other conductive structures (not depicted) that fanout electrical pathways from the conductive pads of the semiconductor chip **100** to package interconnects **2108** (e.g., solder balls).

(77) In some examples, the semiconductor chip **100** and conductive pillars **120** are encased in one or more underfill or mold layers **2110**. It should be noted that the width of the base portion **122** reduces stress on the ELK dielectric material in the semiconductor chip **100**, while the smaller width of the end portion **124** provides greater separation between adjacent solder structures **126** to mitigate against solder bridging. The reduction of the ELK stress mitigates against delamination. Also, there is no need to reduce the size of the solder mask apertures that expose the bond pad **2104** to reduce ELK dielectric material stress as the base portion width is sufficient to reduce the stress on the ELK dielectric material.

(78) For further explanation, FIG. **22** sets forth an example method of fabricating stepped conductive pillar interconnects on a semiconductor chip in accordance with some implementations. The example method of FIG. **22** includes providing **2202** a semiconductor chip substrate including a conductive pad and a passivation layer through which the conductive pad is at least partially exposed. In some implementations, providing **2202** the semiconductor chip substrate is carried out by placing a semiconductor chip substrate (e.g., the semiconductor chip substrate **300** in FIG. **3**) on a carrier (e.g., the carrier **302**). An example of the semiconductor chip substrate includes a device layer implementing integrated circuit devices such as transistor, capacitors, and the like. The semiconductor chip substrate also includes an interconnect layer composed of layers of metallization and dielectric material. The layers of metallization include conductive structures such as conductive traces, pads, and vias interspersed in the dielectric material. In some examples, the dielectric material is ELK or ultra low-k (ULK) dielectric material having a k-value less than 3. Multiple conductive pads are formed in a final layer of metallization on the active surface of the example semiconductor chip substrate. A passivation layer overlays the active surface of the example semiconductor chip substrate. The passivation layer includes apertures over the conductive pads that at least partially expose the conductive pads. In some examples, the

semiconductor chip substrate is attached to the carrier by an adhesive film or through other well-known techniques.

(79) The method of FIG. 22 also includes fabricating **2204** an under-bump metallization layer over an exposed portion of the conductive pad. Such fabrication **2204** is carried out by depositing a layer of metal that acts as an adhesion layer to support the conductive pillar. In some examples, the metal of the adhesion layer is titanium (Ti). The adhesion layer can also act as a diffusion barrier. In some implementations, fabricating **2204** the under-bump metallization layer is also carried out by depositing a layer of metal that acts a seed layer to wet the adhesion layer for fabrication of the conductive pillar. In some examples, the metal of the seed layer is copper (Cu). After fabrication of the conductive pillar is complete, the adhesion layer and seed layer are etched away except for underneath the conductive pillar. The adhesion layer forms the under-bump metallization layer (e.g., the under-bump metallization layer **212** in FIG. 2). A chemical etching can be performed to remove portions of the seed layer and adhesion layer. In some implementations, fabricating **2204** the under-bump metallization layer is carried out as shown and described with references to FIGS. 5 and 13. In some examples, the metal of the adhesion layer and seed layer is deposited on a polymer (e.g., polyimide) layer formed on the passivation layer. The metal of the adhesion layer and the metal of the seed layer are electrically coupled to the conductive pad.

(80) The method of FIG. 22 also includes fabricating **2206** a first portion of the conductive pillar over the conductive pad. The first portion of the conductive pillar has a first width. Fabricating **2206** the first portion is carried out by creating a base portion of the conductive pillar on the seed layer of metal above the conductive pad. The base portion can be fabricated into a variety of shapes, including a circular cylinder, an elliptical cylinder, a cuboid shape, a polygonal cylinder (such as an octagonal cylinder), and the like. In some examples, the base portion is fabricated by electroplating metal within a cavity defined by a resist layer formed on the surface of the seed layer. In some examples, a photolithographic process is employed to create the cavity. In some implementations, fabricating **2206** the first portion of the conductive pillar over the conductive pad is carried out as shown and described with reference to FIGS. 6-8.

(81) The method of FIG. 22 also includes fabricating **2208** a second portion of the conductive pillar on the first portion. The second portion has a second width that is smaller than the first width. Fabricating **2208** the second portion is carried out by creating a base portion of the conductive pillar on the seed layer of metal above the conductive pad. The base portion can be fabricated into a variety of shapes, including a circular cylinder, an elliptical cylinder, a cuboid shape, and the like. In some examples, the base portion is fabricated by electroplating metal within a cavity defined by a resist layer formed on the surface of the seed layer. In some examples, a photolithographic process, such as described below, is employed to create the cavity. In some examples, fabricating **2208** the second portion of the conductive pillar on the first portion is carried out as shown and described with reference to FIGS. 9-11.

(82) For further explanation, FIG. 23 sets forth variations of fabricating stepped conductive pillar interconnects on a semiconductor chip in accordance with some implementations. In the example of FIG. 23, fabricating **2206** the first portion of the conductive pillar over the conductive pad includes dispensing **2302** a first layer of photoresist material. In some examples, dispensing **2302** a first layer of photoresist material is carried out by spin coating photoresist material on the seed layer.

(83) In the example of FIG. 23, fabricating **2206** the first portion of the conductive pillar also includes creating **2304** a first cavity in the first layer of photoresist material by photolithography. In some implementations, creating **2304** a first cavity in the first layer of photoresist material is carried out by exposing a portion of the photoresist material over the conductive pad to radiation (e.g., UV light) and developing the photoresist material to remove the portion of the photoresist material that was exposed.

(84) In the example of FIG. 23, fabricating **2206** the first portion of the conductive pillar also

includes forming **2306** the first portion of the conductive pillar by electroplating metal in the first cavity. In some implementations, forming **2306** the first portion of the conductive pillar is carried out by electroplating metal such as copper on the seed layer within the cavity in the photoresist material.

(85) In the example of FIG. **23**, fabricating **2208** a second portion of the conductive pillar includes dispensing **2308** a second layer of photoresist material. In some implementations, dispensing **2308** a second layer of photoresist material is carried out by spin coating photoresist material on the first layer of photoresist material and the fabricated base portion of the conductive pillar.

(86) In the example method of FIG. **23**, fabricating **2208** a second portion of the conductive pillar on the first portion also includes creating **2310** a second cavity in the second layer of photoresist material by photolithography. The second cavity has a smaller width than the first cavity. In some implementations, creating **2310** the second cavity is carried out by exposing a portion of the second layer of photoresist material over the conductive pad to radiation (e.g., UV light), developing the photoresist material to remove the portion of the photoresist material that was exposed.

(87) In the example method of FIG. **23**, fabricating **2208** a second portion of the conductive pillar on the first portion also includes forming **2312** the second portion of the conductive pillar in the second cavity. In some implementations, forming **2312** the second portion of the conductive pillar in the second cavity is carried out by electroplating metal such as copper on top surfaces of the base portion and the first layer of photoresist material that forms the floor of the cavity in the photoresist material.

(88) In view of the foregoing, it will be appreciated that a number of advantages are realized by stepped conductive pillar interconnects on a semiconductor chip. The smaller width of the end portion of the stepped conductive pillar allows for a fine pitch among the solder joints while mitigating the occurrence solder bridging. The greater width of the base portion of the stepped conductive pillar allows for a wider UBM layer that mitigates ELK dielectric layer stress, thus mitigating the occurrence of cracking or delamination in the semiconductor chip.

(89) It will be understood from the foregoing description that modifications and changes can be made in various implementations of the present disclosure. The descriptions in this specification are for purposes of illustration only and are not to be construed in a limiting sense. The scope of the present disclosure is limited only by the language of the following claims.

Claims

1. A semiconductor chip comprising: a device layer; an interconnect layer fabricated on the device layer, the interconnect layer including a conductive pad disposed on a final dielectric layer of the interconnect layer; a passivation layer formed on the interconnect layer; a conductive pillar coupled to the conductive pad and including at least a first portion having a first width and a second portion having a second width, the first portion being disposed between the second portion and the conductive pad, wherein the first width of the first portion is greater than the second width of the second portion, and a third portion disposed between the first portion and the second portion, the third portion having a third width that is smaller than the second width; an under-bump metallization layer formed on the conductive pad and coupling the conductive pillar to the conductive pad; and a polymer layer formed on the passivation layer and between the passivation layer and the under-bump metallization layer.
2. The semiconductor chip of claim 1 further comprising a solder cap on an end of the second portion.
3. The semiconductor chip of claim 1 further comprising: an aperture through the passivation layer and the polymer layer, the aperture exposing a portion of the conductive pad, wherein the under-bump metallization layer is formed on at least the exposed portion of the conductive pad.
4. The semiconductor chip of claim 1 further comprising: a passivation layer formed on the

interconnect layer; an aperture in the passivation layer, the aperture exposing a portion of the conductive pad; an under-bump metallization layer formed on at least the exposed portion of the conductive pad and coupling the conductive pillar to the conductive pad; and a polymer layer formed on the first portion of the conductive pillar and the under-bump metallization layer.

5. The semiconductor chip of claim 1 further comprising: a passivation layer formed on the interconnect layer; a first polymer layer formed on the passivation layer; an aperture in the passivation layer and the first polymer layer, the aperture exposing a portion of the conductive pad; an under-bump metallization layer formed on at least the exposed portion of the conductive pad and coupling the conductive pillar to the conductive pad; and a second polymer layer formed on the first portion of the conductive pillar and the under-bump metallization layer.

6. The semiconductor chip of claim 1, wherein the conductive pillar forms a stepped cylinder.

7. The semiconductor chip of claim 1, wherein the conductive pillar forms a stepped cuboid.

8. The semiconductor chip of claim 1, wherein the interconnect layer includes extremely low-k (ELK) dielectric material.

9. A semiconductor chip comprising: a device layer; an interconnect layer fabricated on the device layer, the interconnect layer including a conductive pad; and a conductive pillar coupled to the conductive pad and including at least a first portion having a first width and a second portion having a second width, the first portion being disposed between the second portion and the conductive pad, wherein the first width of the first portion is greater than the second width of the second portion, wherein the conductive pillar forms a stepped elliptical cylinder, wherein the first portion and the second portion each include opposing identical elliptical faces separated by a cylindrical sidewall, wherein the conductive pillar further includes a third portion disposed between the first portion and the second portion, the third portion having a third width that is smaller than the second width.

10. A semiconductor chip comprising: a device layer; an interconnect layer fabricated on the device layer, the interconnect layer including a conductive pad; and a conductive pillar coupled to the conductive pad and including at least a first portion having a first width and a second portion having a second width, the first portion being disposed between the second portion and the conductive pad, the second portion being offset from the first portion, wherein the first width of the first portion is greater than the second width of the second portion, wherein the first portion has an axis that is off-center with respect to an axis of the second portion, and wherein the conductive pillar further includes a third portion disposed between the first portion and the second portion, the third portion having a third width that is smaller than the second width.

11. The semiconductor chip of claim 10 further comprising a solder cap on an end of the second portion.

12. The semiconductor chip of claim 10 further comprising: a passivation layer formed on the interconnect layer; a polymer layer formed on the passivation layer; an aperture through the passivation layer and the polymer layer, the aperture exposing a portion of the conductive pad; and an under-bump metallization layer formed on at least the exposed portion of the conductive pad and coupling the conductive pillar to the conductive pad.

13. The semiconductor chip of claim 10 further comprising: a passivation layer formed on the interconnect layer; an aperture in the passivation layer, the aperture exposing a portion of the conductive pad; an under-bump metallization layer formed on at least the exposed portion of the conductive pad and coupling the conductive pillar to the conductive pad; and a polymer layer formed on the first portion of the conductive pillar and the under-bump metallization layer.

14. The semiconductor chip of claim 10 further comprising: a passivation layer formed on the interconnect layer; a first polymer layer formed on the passivation layer; an aperture in the passivation layer and the first polymer layer, the aperture exposing a portion of the conductive pad; an under-bump metallization layer formed on at least the exposed portion of the conductive pad and coupling the conductive pillar to the conductive pad; and a second polymer layer formed on the

first portion of the conductive pillar and the under-bump metallization layer.

15. The semiconductor chip of claim 10, wherein the conductive pillar forms a stepped elliptical cylinder, wherein the first portion and the second portion each include opposing identical elliptical faces separated by a cylindrical sidewall.

16. The semiconductor chip of claim 10, wherein the conductive pillar forms a stepped cylinder having the offset.

17. The semiconductor chip of claim 9 further comprising a solder cap on an end of the second portion.
